The Oracle Architecture

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The Oracle Architecture

- Oracle is based on the **client–server** architecture.
- The Oracle server consists of the **database** (the raw data, including log and control files) and the **instance** (the processes and system memory on the server that provide access to the database).
- An instance can connect to only one database. The database consists of a logical structure, such as the database schema, and a physical structure, containing the files that make up an Oracle database.

Oracle’s logical database structure

- It is defined as how storage is allocated and organized, independent of physical files on disk.
- This hierarchy allows the database to manage space efficiently and provides a clear separation from the operating system’s physical view.
- At the logical level, Oracle maintains tablespaces, schemas, data blocks and extents/segments.

Data blocks

- The data block is the smallest unit of storage that Oracle can use or allocate.
- One data block corresponds to a specific number of bytes of physical disk space.
- The data block size can be set for each Oracle database when it is created.
- This data block size should be a multiple of the operating system's block size (within the system's maximum operating limit) to avoid unnecessary I/O.
- A data block has the following structure:
- ***Header*** Contains general information such as block address and type of segment.

- ***Table directory*** Contains information about the tables that have data in the data block.
- • ***Row directory*** Contains information about the rows in the data block.
- • ***Row data*** Contains the actual rows of table data. A row can span blocks.
- • ***Free space*** Allocated for the insertion of new rows and updates to rows that require additional space.

Extent

- The next level of logical database space is called an **extent**.
- An extent is a specific number of contiguous data blocks allocated for storing a specific type of information

Segment

- The level above an extent is called a **segment.**
- A segment is a set of extents allocated for a certain logical structure.
- For example, each table's data is stored in its own data segment, while each index's data is stored in its own index segment.
- Oracle dynamically allocates space when the existing extents of a segment become full.
- Because extents are allocated as needed, the extents of a segment may or may not be contiguous on disk

Tablespaces

- An Oracle database is divided into logical storage units called **tablespaces**. A tablespace is used to group related logical structures together. For example, tablespaces commonly group all the application’s objects to simplify some administrative operations.
- Every Oracle database contains a tablespace named SYSTEM, which is created automatically when the database is created. The SYSTEM tablespace always contains the system catalog tables (called the data dictionary in Oracle) for the entire database. A small database might need only the SYSTEM tablespace;
- however, it is recommended that at least one additional tablespace is created to store user data separate from the data dictionary, thereby reducing contention among dictionary objects and schema objects for the same datafiles.

User and A Schema

- A user (sometimes called a username) is a name defined in the database that can connect to, and access, objects.
- A schema is a named collection of schema objects, such as tables, views, indexes, clusters, and procedures, associated with a particular user

Oracle's physical database structure

- It consist of files on a disk that stores data, visible at the operating system level
- The main physical database structures in Oracle are datafiles, redo log files, and control files.
- **Datafiles**
- Every Oracle database has one or more physical **datafiles.** The data of logical database structures (such as tables and indexes) is physically stored in these datafiles. The simplest Oracle database would have one tablespace and one datafile.
- A more complex database might have four tablespaces, each consisting of two datafiles, giving a total of eight datafiles.

Redo log files

- Every Oracle database has a set of two or more redo log files that record all changes made to data for recovery purposes.
- Should a failure prevent modified data from being permanently written to the datafiles, the changes can be obtained from the redo log, thus preventing work from being lost.

Control files

- Every Oracle database has a control file that contains a list of all the other files that make up the database, such as the datafiles and redo log files.
- For added protection, it is recommended that the control file should be multiplexed (multiple copies may be written to multiple devices).
- Similarly, it may be advisable to multiplex the redo log files as well.

The Oracle instance

- The Oracle instance consists of the Oracle processes and shared memory required to access information in the database. The instance is made up of the Oracle background processes, the user processes, and the shared memory used by these processes.
- Oracle uses shared memory for caching data and indexes as well as storing shared program code.
- Shared memory is broken into various memory structures, of which the basic ones are the System Global Area (SGA) and the Program Global Area (PGA)

System Global area

- System global area The SGA is an area of shared memory that is used to store data and control information for one Oracle instance. The SGA is allocated when the Oracle instance starts and deallocated when the Oracle instance shuts down
- Database buffer cache This contains the most recently used data blocks from the database.
- These blocks can contain modified data that has not yet been written to disk (dirty blocks), blocks that have not been modified, or blocks

- ***Database buffer cache*** This contains the most recently used data blocks from the database.
- These blocks can contain modified data that has not yet been written to disk (dirty blocks)
- **Redo log buffer** This contains the redo log file entries, which are used for recovery purposes.
- The background process LGWR writes the redo log buffer to the active online redo log file on disk.
- **Shared pool** This contains the shared memory structures such as shared SQL areas in the library cache and internal information in the data dictionary. The shared SQL areas contain parse trees and execution plans for the SQL quer

Program global area

- Program global area The PGA is an area of shared memory that is used to store data and control information for the Oracle server processes. The size and content of the PGA depends on the Oracle server options installed.
- ***User processes*** Each user process represents the user’s connection to the Oracle server (for example, through SQL*Plus or an Oracle Forms application).

- **Oracle processes** Oracle (server) processes perform functions for users. Oracle processes can be
- split into two groups: server processes (which handle requests from connected user processes) and background processes
- **Database Writer (DBWR)** The DBWR process is responsible for writing the modified (dirty) blocks from the buffer cache in the SGA to datafiles on disk.
- **Log Writer** (LGWR) The LGWR process is responsible for writing data from the log buffer to the redo log.
- **CHKPT CKPT** was responsible for signaling regular checkpoints.

PMON

- PMON monitors all the server processes and detects any problems with the sessions. If a session has terminated abnormally, PMON
- **SMON** mounts a database by locating and validating the database controlfile. It then opens a database by locating and validating all the datafiles and online log files.